

INDIAN STATISTICAL INSTITUTE

Practice Problems, August 13, 2026
M.TECH(CS/CrS) I YEAR

INTRODUCTION TO PROGRAMMING

1. The file `inps.dat` contains student records. The first line specifies the number of records, and each of the subsequent lines contains a student's name, roll number, and marks. Perform the following tasks:
 - (a) Define a structure with fields for the student's name, roll number, and marks, and create an array of such structures.
 - (b) Write a routine to populate the array by reading input from `stdin` in the same format as `inps.dat`.
 - (c) Write a routine to print the contents of the array in the format of `inps.dat`.
 - (d) Sort the array based on each of the three fields using any sorting algorithm of your choice.
2. Repeat all of the above tasks assuming that the array stores pointers to structures instead of structures themselves.